Abhinav Kumar

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Summary

Aspiring Computer Science (AI & ML) undergraduate and Full-Stack Developer with hands-on experience building MERN applications and RESTful APIs. Skilled in backend architecture, database integration, system optimization, and problem-solving, with a focus on developing scalable and reliable web applications. Seeking a backend-focused software engineering role to build robust, production-ready systems.

Education

RNS Institute of Technology — **Bengaluru, India**
Bachelor of Engineering – Computer Science (AI & ML) — *Sep 2023 – June 2027*

- Current GPA: 8.41/10 | 7th semester
- Relevant coursework: Data Structures & Algorithms, Object-Oriented Programming, Operating Systems, Computer Networks, Database Management Systems, Machine Learning

Technical Skills

Languages: C, C++, Java, Python, JavaScript, TypeScript
Frontend: React.js, Tailwind CSS, HTML5, CSS3, Chart.js
Backend: Node.js, Express.js, REST API Development, MVC Architecture, JWT Authentication, bcrypt.js, js-tiktoken
Databases: MongoDB, MySQL, Firebase Firestore
Tools & Platforms: Git, GitHub, Postman, Firebase, VS Code
CS Fundamentals: Data Structures & Algorithms (DSA), OOP, OS, Computer Networks, DBMS

Projects

Prompt Optimizer | *Node.js, Express.js, MongoDB, React.js, Tailwind CSS, js-tiktoken, JWT*

- Built a full-stack prompt optimization platform that removes conversational filler, redundant phrases, and formatting noise through a rule-based text compression pipeline while preserving core instructions.
- Integrated `js-tiktoken` for token-level analysis and developed a multi-provider pricing engine to estimate prompt costs across OpenAI, Anthropic, and Google models.
- Designed RESTful APIs using MVC architecture with JWT authentication, bcrypt password hashing, and MongoDB-based per-user prompt optimization history.
- Enabled users to compare original vs. optimized prompts with token savings, compression percentage, and estimated cost reduction before sending prompts to LLM APIs.

KalaSetu – AR-Based Cultural Commerce Platform | *React.js, Tailwind CSS, Node.js, Express.js, MongoDB, AR.js*

- Developed an augmented reality marketplace enabling users to visualize traditional crafts and decor within real-world environments before purchase.
- Designed a digital preservation module that archives indigenous patterns, techniques, and artistic styles for long-term accessibility.
- Built a direct artisan-to-consumer platform eliminating intermediaries and enabling rural creators to manage storefronts and reach global buyers.
- Implemented inclusive UI/UX architecture ensuring usability for individuals with varying levels of digital literacy.

Email Scheduler – Distributed Task Processing System | *Node.js, Kafka, Redis, PostgreSQL, Prisma*

- Built a distributed email scheduler using **Kafka**, **Redis**, and **PostgreSQL** for event-driven processing, delayed scheduling, and reliable multi-worker execution.
- Implemented **Redis locks and sorted sets** for atomic job claiming and used **Kafka partitioning by job ID** to prevent duplicate processing and maintain event ordering.
- Managed **PENDING → PROCESSING → SENT/FAILED** states with Prisma/PostgreSQL and implemented a **dead-letter queue** for failed-job isolation and retries.

Achievements

- Solved **450+ problems** across competitive programming platforms, strengthening expertise in data structures, algorithms, and problem-solving.
- Achieved a **2⋆ rating on CodeChef**, **1000+ rating on Codeforces**, and **1550+ rating on LeetCode**, demonstrating consistent competitive programming and problem-solving skills.
- Secured a **Semi-Finalist** position among **2,000+ teams** in **PW-RIFT**, a 24-hour national-level hackathon, by developing an innovative technical solution under strict time constraints.